Automated Data ETL + Validation System (Team 9)

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Deliverable 2 (D2)

1. Situation

The system under consideration is the dataset handling workflow of college students working on data science and machine learning projects, as defined in D1. Students select datasets, upload them (mostly CSV/Excel), and manually perform cleaning and preprocessing before model training. Observations indicate variation in expertise and inconsistency in dataset quality, leading to time overhead and impact on model accuracy.

2. Constraints

- Time constraints: Data cleaning consumes significant time before model training.
- User capability variation: Students show different levels of preprocessing confidence.
- Data format limitations: Mostly CSV/Excel; some uncertainty about file types.
- Data quality variability: Datasets often noisy, incomplete, or inconsistent.
- Manual workflow dependency: Manual handling introduces inefficiencies.

3. Observed Pain Points

- Data quality inconsistency: Noisy, incomplete, poorly structured datasets.
- Time overhead in preprocessing before model training.
- Knowledge gap regarding dataset formats and preprocessing steps.
- Difficulty in finding relevant and suitable datasets.
- Poor-quality data affecting model accuracy and reliability.

4. Prioritisation and Focus Selection

- Candidate Focus Areas Considered:
  Data Quality Inconsistency
  Dataset Selection Difficulty
- Selected Primary Focus:
  Data Quality Inconsistency, due to its direct impact on model performance, time overhead, and interaction with other pain points.

5. Evidence Strength and Limitations

- Evidence is based on structured survey responses from 53 students. Multiple responses highlight noisy data, preprocessing difficulty, and impact on model accuracy.
- Limitations include self-reported bias, limited sample scope, and lack of quantitative error measurement.

6. Assumptions and Risks

- Data quality issues significantly contribute to inefficiency.
- Students lack structured mechanisms to detect inconsistencies.
- Improving dataset reliability may improve model performance.

Risks include misidentifying the root cause (skill-based vs. data-based issues) and overestimating the impact of data quality alone.

7. Scope of Further Work

- To Be Explored:

  Patterns of dataset inconsistencies, types of common data issues, and their impact on workflow.

- Out of Scope:

  Solution design, feature definition, architecture planning, and prototyping.

8. Direction and Tentative Intent

Future work will explore structured mechanisms to improve reliability in dataset preparation workflows.
Tentative Intent: Product-oriented.

9. Summary

This D2 selects Data Quality Inconsistency as the primary focus based on D1 findings. The choice is justified by its frequency, impact on model performance, and interaction with other workflow inefficiencies.